

**CE586 EARTHQUAKE ENGINEERING
2025-2026 SPRING SEMESTER**

ASSIGNMENT #3 (DUE 4 MAY)

Q1. Using the ground motion records assigned to you in below table, obtain and plot (between periods of 0.2 seconds and 3.0 seconds with a period increment of 0.1 seconds) the below listed response spectra for linear elastic SDOF systems subjected to the assigned ground motion record. You are expected to write your own code based on the Newmark-Beta algorithm provided in the lecture notes (you can use any programming language or platform) or, as the second alternative, you can use readily available, openware, research-oriented software (OpenSEES, Seismosignal, USDP, II-DAP, PRISM etc.) to carry out dynamic analyses of SDOF structures. If you write a code, please present it with your homework report.

- a)** Displacement spectra for $\xi=2\%$, 5% and 10% (on the same graph).
- b)** Pseudo acceleration spectra for $\xi=2\%$, 5% and 10% (on the same graph).
- c)** Acceleration spectra for $\xi=2\%$, 5% and 10% (on the same graph).
- d)** Comment on the accuracy of pseudo terms (for acceleration) when compared to the actual spectral values. You can show pseudo and actual spectral terms on the same graph for comparison.

Q2. Imagine that you are the structural designer in a firm and you are going to design a new mid-rise RC residential building which is very close to the recording station of the ground motion record assigned to you (see the coordinates from below table together with the soil class assigned to each location).

- a)** Obtain 5% damped elastic design spectra of this building for a return periods of 475 and 2475 years by using the 2007 Turkish Seismic Code (make reasonable assumptions for the seismic zone and site class by using the data in the below table).
- b)** Obtain 5% damped elastic design spectra of this building for a return periods of 475 and 2475 years by using the 2018 Turkish Seismic Code.
- c)** Compare the 5% damped elastic acceleration response spectrum with the design spectra that you obtained in part (a) and (b) by putting them on the same plot. Discuss briefly.
- d)** Obtain the inelastic response spectrum for constant R-factor by using the elastic acceleration spectra with 5% damping ratio (without using numerical analysis) for R values of 4 and 8. Then obtain the corresponding inelastic design spectra for R=4 and 8 in the context of both 2007 Turkish Seismic Code and 2018 Turkish Seismic Code. Compare them on the same plot (separately for R=4 and R=8). For any missing data, you can make assumptions by stating their reasons. Discuss the results briefly.

Notes:

1) You have to register to “e-Devlet” application to have an access to website of the 2018 Turkish Earthquake Hazard Map (<https://tdth.afad.gov.tr/>) for obtaining the relevant information for your homework question.

2) The code of the ground motion (GM) records provided for each student is listed below. Ground motion data is given in two columns: first one, time increment (in seconds), second one, ground acceleration (in cm/s^2). Additional information regarding the earthquake of the assigned record can be obtained from the Excel table “Ground Motion List” in the zip file which also contains the digital data of the records.

Student ID	GM Code	Latitude	Longitude	Soil Class
2215879	HOR83X01	40.037	42.157	ZC
2370393	ERZ92XY01	39.579	39.687	ZC
2735835	DIN95Y01	38.247	30.49	ZC
2529600	MAR99X02	40.774	29.932	ZC
2743862	MAR99X04	40.885	30.905	ZE
2369916	DZC99X02	40.875	31.228	ZD
2487866	PZR23Y01	37.487	37.298	ZE
1276997	PZR23Y02	37.502	37.364	ZE
2137644	NDG23Y03	37.184	36.751	ZE
2161438	NDG23Y05	37.197	36.727	ZC
2370088	ELB23Y06	38.222	37.173	ZC
2782134	ELB23Y07	38.188	37.227	ZD
2416055	PZR23Y08	37.512	37.309	ZD